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Electropneumatic valve assembly, electropneumatic brake system, commercial vehicle

Abstract

An electropneumatic valve assembly is provided for actuating a parking brake function of an electropneumatic brake system of a commercial vehicle. The assembly includes a parking brake valve unit which receives supply pressure via a supply pressure path. A pneumatically self-holding trailer valve unit is connected to the supply pressure path and has a pneumatically switchable trailer supply valve. The supply valve provides a trailer supply pressure to a trailer supply port in dependence on an electronic trailer brake signal. The parking brake valve unit has a pilot valve unit and a main valve unit and the pilot valve unit outputs a parking brake control pressure at the main valve unit. The main valve unit outputs a parking brake pressure at a spring brake port. The valve assembly has a compensation valve to maintain the parking brake control pressure outputted at the main valve unit to compensate for valve leakage.

Inventors:	van Thiel; Julian (Grossburgwedel, DE)
Applicant:	ZF CV Systems Global GmbH (Bern, CH)
Family ID:	84533939
Assignee:	ZF CV Systems Global GmbH (Bern, CH)
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Primary Examiner: Williams; Melanie Torres

Attorney, Agent or Firm: Walter Ottesen, P.A.

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

(1) This application claims priority of German patent application no. 10 2021 117 460.0, filed Jul. 6, 2021, the entire content of which is incorporated herein by reference.

TECHNICAL FIELD

(2) The disclosure relates to an electropneumatic valve assembly. The disclosure also relates to an electropneumatic brake system having the electropneumatic valve assembly and to a commercial vehicle having the electropneumatic brake system.

BACKGROUND

(3) In modern electropneumatic brake systems, in particular in parking brake systems, it is important that the brake system behaves reliably. This applies both to normal operation of the commercial vehicle and also in the event of a fault in the commercial vehicle, in particular in the electropneumatic brake system.

(4) In particular in the event of leakages in the electropneumatic brake system, there is a risk that unintentional and/or uncontrolled switching operations will occur.

(5) DE 10 2016 011 390 A1 describes an electropneumatic parking brake valve device with three pneumatic ports, which form a compressed air input for connection to a compressed air supply, a compressed air output for connection to the spring parking brake, and an air release port, wherein there is provided in the electropneumatic parking brake valve device, between a control volume and a compressed air input and/or an air release port, a bypass line having a bypass valve device which is pneumatically actuated by the control volume for connecting or separating the control volume to or from a port, and in at least one of the steady-state positions the bypass valve device is actuated by the control volume such that it at least partially bypasses an inlet and outlet valve device for admitting air to the control volume and connects the control volume to the port.

(6) On the one hand, a leakage can, for example, lead to an uncontrolled pressure drop in a pilot branch of a main valve unit in the air admission state, but on the other hand it can also lead to an uncontrolled pressure build-up in the pilot branch in an air release state. Leakages can occur in particular in the case of articulated trucks, where the pneumatic system of a commercial vehicle is extended via at least one separable pneumatic connection to a trailer.

(7) Electropneumatic brake systems are in need of further improvement, in particular with regard to reliable behavior and a reduced risk of uncontrolled operation. This applies in particular to electropneumatic parking brake systems in commercial vehicles and in particular in articulated trucks.

(8) It is therefore desirable to improve the functioning of the electropneumatic valve assembly for an electropneumatic brake system.

SUMMARY

(9) An object of the disclosure is to provide an improved electropneumatic valve assembly. In particular, there is to be provided an electropneumatic valve assembly which even in the event of leakages ensures reliable, controlled behavior of a parking brake function of the electropneumatic brake system.

(10) The disclosure proceeds from an electropneumatic valve assembly for actuating a parking brake function of an electropneumatic brake system of a commercial vehicle, having a parking brake valve unit which receives supply pressure from at least one pressure supply via a supply pressure path, and having a pneumatically self-holding trailer valve unit which is connected to the supply pressure path for receiving the supply pressure and has a pneumatically switchable trailer supply valve which provides a trailer supply pressure to a trailer supply port in dependence on an electronic trailer brake signal, wherein the parking brake valve unit has a pilot valve unit and a main valve unit, and the pilot valve unit is provided to output a parking brake control pressure at the main valve unit, wherein the main valve unit is configured to output a parking brake pressure at least at one spring brake port in dependence on the received parking brake control pressure.

(11) According to the disclosure, it is provided in the electropneumatic valve assembly according to the first aspect of the disclosure that the electropneumatic valve assembly has a compensation valve which is provided to maintain the parking brake control pressure outputted at the main valve unit by the pilot valve unit and thus to compensate at least partially for a leakage of the pilot valve unit and/or the main valve unit.

(12) Via the compensation valve according to the disclosure, leakages which occur in the pilot valve unit and/or the main valve unit can advantageously be compensated for. This is advantageously made possible by a controllable pneumatic connection to the at least one pressure

supply or to the air release port, which can be established by the compensation valve. Because the trailer valve unit is pneumatically connected to the supply pressure path for receiving the supply pressure, a trailer that is pneumatically connected to the commercial vehicle is advantageously included in the compensation and thus likewise secured.

(13) In an advantageous further embodiment of the disclosure it is provided that the compensation valve is configured to compensate at least partially for a leakage of the pilot valve unit in the air admission state and in the air release state.

(14) The invention is developed further in that the compensation valve is in the form of a 3/2-way valve and has a first compensation valve port which receives supply pressure, a second compensation valve port at which the parking brake control pressure or a pressure derived therefrom can be present, and a third compensation valve port which is connected to an air outlet, wherein in a first switch position the second compensation valve port is connected to the third compensation valve port, and in a second switch position the first compensation valve port is connected to the second compensation valve port.

(15) It is advantageously provided that the first compensation valve port is connected to the supply pressure path. It is advantageously provided that the second compensation valve port is connected to a line which carries the parking brake control pressure, in particular a control line. By switching into the second switch position, a pneumatic connection between a control line and an air release port can advantageously be established.

(16) In an advantageous further embodiment of the disclosure it is provided that the compensation valve is biased into the first switch position by spring loading and switches into the second switch position in the actuated state. In particular, the compensation valve has a return spring.

(17) The invention is developed further in that the compensation valve is in the form of a pneumatically switchable valve which is switchable via a compensation valve control pressure. The compensation valve control pressure is advantageously provided at a compensation valve control port.

(18) In an advantageous further embodiment of the disclosure it is provided that the compensation valve is configured to switch when the compensation valve control pressure exceeds a compensation valve threshold value. The compensation valve is advantageously configured, in particular by the configuration of a return spring, to switch into the second switch position when the compensation valve threshold value is exceeded. The compensation valve threshold value is advantageously between 0.241 MPa (35 psi) and 0.379 MPa (55 psi), preferably between 0.276 MPa (40 psi) and 0.345 MPa (50 psi), particularly preferably 0.310 MPa (45 psi).

(19) In an advantageous further embodiment of the disclosure it is provided that the compensation valve control pressure is a pressure that is present or outputted at the second compensation valve port, whereby pneumatic self-holding for the compensation valve is implemented. It is advantageously provided that the second compensation valve port is pneumatically connected to the compensation valve control port via a compensation control path.

(20) It is further achieved via the compensating pneumatic connection to the at least one pressure supply that can be established by the compensation valve that a pilot pressure of the main valve unit can be coupled with the supply pressure in the at least one pressure supply. Via the pneumatic connection between the supply pressure path and the trailer valve unit, the above-described pneumatic coupling can also comprise a pneumatic system of a trailer connected to the trailer valve unit. If the compensation valve control pressure falls below a compensation valve threshold value, the self-holding nature leads to switching of the compensation valve into the first switch position, whereby the at least one parking brake cylinder engages reliably and reproducibly, namely when the compensation valve threshold value is not reached.

(21) It is also advantageously achieved via the compensation control path that, in the event of pumping down or a purposive pressure drop in at least one, in particular all, of the compressed air supplies and an associated fall in the compensation valve control pressure below a compensation

valve threshold value, switching into the first switch position takes place in order to release air from the control line and thus activate a parking brake function.

(22) In an advantageous further embodiment of the disclosure it is provided that the compensation valve control pressure is a parking brake pressure that is present or outputted at the spring brake port. In such a further embodiment, the compensation valve is thus switchable in dependence on the outputted parking brake pressure. It is advantageously provided that the spring brake port is pneumatically connected to the compensation valve control port via a compensation control path in the form of a feedback path.

(23) In an advantageous further embodiment of the disclosure it is provided that the compensation valve is throttled, preferably such that the connection of the first compensation valve port to the second compensation valve port and/or the connection of the second compensation valve port to the third compensation valve port is in each case throttled. In particular, the compensation valve has a first throttle for throttling the pneumatic connection between the first compensation valve port and the second compensation valve port in the second switch position. In particular, the compensation valve has a second throttle for throttling the pneumatic connection between the second compensation valve port and the third compensation valve port in the first switch position. In other further embodiments, a common throttle can be provided, which is independent of the switch position of the compensation valve and is connected upstream or downstream of the compensation valve. Such a common throttle can advantageously be arranged between the second compensation valve port and a branch node for the compensation control path.

(24) The invention is developed further in that the compensation valve is arranged in a compensation path, wherein the compensation path pneumatically connects the supply pressure path to a control line, and the control line, for providing the parking brake control pressure, is pneumatically connected to the main valve unit, preferably to a control port of the main valve unit.

(25) In an advantageous further embodiment of the disclosure there is provided a deactivation valve which is arranged in the compensation path and has a first deactivation valve port and a second deactivation valve port and which, in dependence on a deactivation signal, can be switched from an activation position which pneumatically connects the first deactivation valve port and the second deactivation valve port into a deactivation position which pneumatically separates the first deactivation valve port and the second deactivation valve port.

(26) The deactivation valve is advantageously in the form of a 2/2-way valve, preferably in the form of a 2/2-way solenoid valve. The deactivation valve is advantageously arranged between the compensation valve and the control line. Via a deactivation valve, the pneumatic connection established by the compensation valve can advantageously be switchable and in particular can be blocked for a short time. This is advantageous if precisely timed actuation, in particular via the pilot valve unit, is required. Switching into the deactivation position prevents a pressure drop via the compensation valve, which improves the control quality in particular in the case of an actuation of the main valve unit in a stepped manner and/or with high temporal resolution via the control line. Switching into the deactivation position can advantageously also reduce air consumption.

(27) In an advantageous further embodiment of the disclosure, a trailer pilot unit is provided for providing a trailer control pressure in dependence on the electronic trailer brake signal, wherein the trailer supply valve is pneumatically actuatable in dependence on the trailer control pressure.

(28) Within the context of a preferred further embodiment there is provided an emergency release port which can be pneumatically connected to the control port in order to provide an emergency release pressure. Via an emergency release port, the parking brake function, in particular engaged spring brake cylinders, can advantageously be released even if the supply pressure has fallen below the first threshold value. In particular, the emergency release port can be used with a further compressed air source which is independent of the compressed air supply that supplies the parking brake function. Such a compressed air source can in particular be a compressed air source that supplies the service brake system. Via the compensation valve, with a one-time inputting of an

emergency release pressure, the compensation valve can advantageously be switched into the second switch position for admitting air permanently, and thus in particular independently of the emergency release pressure, to the spring brake port and thus the parking brake function.

(29) Within the context of a further embodiment it is provided that there is arranged at the emergency release port a check valve which opens in the direction of the control port and closes in the opposite direction.

(30) The invention is advantageously developed further by an anti-compound port which can be pneumatically connected to the control port in order to provide an additional parking brake pressure.

(31) The invention is advantageously developed further by a selection valve having a first selection valve port which is connected to the anti-compound port, a second selection valve port which is connected to the emergency release port, and a third selection valve port which is connected to the control port, wherein the selection valve is configured to connect to the third selection valve port the selection valve port, from the first selection valve port and the second selection valve port, at which the higher pressure is present.

(32) In a second aspect, the disclosure further provides, for achieving the object, an electropneumatic brake system for a commercial vehicle, having an electropneumatic valve assembly according to the first aspect of the disclosure, at least a first compressed air supply, at least one spring brake cylinder which is pneumatically connected to the at least one spring brake port for receiving the parking brake pressure, and an electronic control unit, configured to provide at least one electronic parking brake signal.

(33) In a third aspect, the disclosure further provides, for achieving the object, a commercial vehicle, having an electropneumatic valve assembly according to the first aspect of the disclosure and/or an electropneumatic brake system according to the second aspect of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The invention will now be described with reference to the drawings wherein:
- (2) FIG. 1 shows a first embodiment of an electropneumatic valve assembly according to the disclosure;
- (3) FIG. 2 shows a second embodiment of an electropneumatic valve assembly according to the disclosure;
- (4) FIG. 3 shows a third embodiment of an electropneumatic valve assembly according to the disclosure; and,
- (5) FIG. 4 shows an electropneumatic brake system having a valve assembly according to the disclosure.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

- (6) FIG. 1 shows a first embodiment of an electropneumatic valve assembly **1** according to the concept of the invention. The electropneumatic valve assembly **1** has a parking brake valve unit **24** with a supply pressure path **70**. The parking brake valve unit **24** is configured to output a parking brake pressure pBP at least at one spring brake port **21** in dependence on an electronic parking brake signal SFB.
- (7) The supply pressure path **70** is pneumatically connected via a supply shuttle valve **49** to a first compressed air supply **6** and a second compressed air supply **7** in order to receive a supply pressure pV. The supply shuttle valve **49** is in the form of a select-high shuttle valve, so that, of the first compressed air supply **6** and the second compressed air supply **7**, the compressed air supply in which the higher supply pressure pV prevails is connected to the supply pressure path **70** and the respective other compressed air supply **6**, **7** is closed.

(8) The supply pressure path **70** is divided into a supply branch **74**, a control branch **76** and a compensation path **86**. The compensation path **86** pneumatically connects the supply pressure path **70** to a control line **36**.

(9) A compensation valve **80** is arranged in the compensation path **86**. The compensation valve **80** is in the form of a 3/2-way valve **81**. The compensation valve **80** has a first compensation valve port **80.1**, which is pneumatically connected to the supply pressure path **70**. The compensation valve **80** has a second compensation valve port **80.2**, which is pneumatically connected to the compensation path **86**. The compensation valve **80** has a third compensation valve port **80.3**, which is pneumatically connected to an air release line **44**. The air release line **44** is connected to an air release port **3**, which releases air into the environment.

(10) The compensation valve **80** has a compensation valve control port **80.4**, which is pneumatically connected via a compensation control path **83** to the second compensation valve port **80.2**. Pneumatic self-holding is preferably implemented via the compensation control path **83**. Via the compensation control path **83**, a pressure present at the second compensation valve port **80.2**, in particular a parking brake control pressure pPS, is provided at the compensation valve control port **80.4**.

(11) In a second switch position **80B** of the compensation valve **80**, the first compensation valve port **80.1** is pneumatically connected to the second compensation valve port **80.2**, and the third compensation valve port **80.3** is preferably closed. In the second switch position **80B**, the supply pressure path **70** is thus pneumatically connected to the compensation path **86** and thus to the control line **36**. In a first switch position **80A** of the compensation valve **80**, the second compensation valve port **80.2** is pneumatically connected to the third compensation valve port **80.3**, and the first compensation valve port **80.1** is preferably closed. In the first switch position **80A**, the compensation path **86** is thus connected to the air release line **44** and the control line **36** is thus pneumatically connected to the air release port **3**.

(12) In the present case, the compensation valve **80** has a first throttle **27** between the first compensation valve port **80.1** and the second compensation valve port **80.2**. The first throttle **27** advantageously has a nominal width which is smaller compared to the compensation path **86** and/or the supply pressure path **70**. In the present case, the compensation valve **80** has a second throttle **28** between the second compensation valve port **80.2** and the third compensation valve port **80.3**. The second throttle **28** advantageously has a nominal width which is smaller compared to the compensation path **86** and/or the air release line **44**.

(13) In the present case, the parking brake valve unit **24** has a main valve unit **34**. The main valve unit **34** comprises a pilot valve unit **32** and a pneumatically actuated main valve **18**. The pilot valve unit **32** is configured to provide a parking brake control pressure pPS at a control port **18.3** of the pneumatically actuated main valve **18** in dependence on the electronic parking brake signal SFB. The pneumatically actuated main valve **18** is configured to output the parking brake pressure pBP, which is provided at the spring brake port **21**, in dependence on this parking brake control pressure pPS at a main port **18.2**. The pneumatically actuated main valve **18** is in the present case in the form of a relay valve **20**.

(14) The pilot valve unit **32** has a first pilot valve **41** and a second pilot valve **42**, which are each in the form of 2/2-way solenoid valves. Via the first pilot valve **41**, the supply pressure pV, in an open position **41A**, can be provided from the supply pressure path **70** as parking brake control pressure pPS at the control line **36**. The control line **36** is arranged between the first pilot valve **41** and the second pilot valve **42** and is pneumatically connected or connectable to the control port **18.3** of the main valve **18**.

(15) By switching the first pilot valve **41** into a closed position **41B**, the pilot pressure pSV can be trapped or held in the control line **36**, in particular at the control port **18.3**, for permanent actuation. The second pilot valve **42** is thereby likewise in a closed position **42B**. Corresponding to the construction of the pilot valve unit **32** with a first pilot valve **41** and a second pilot valve **42**, the

electronic parking brake signal SFB comprises a first electronic parking brake signal SFB1 for actuating the first pilot valve **41** and a second electronic parking brake signal SFB2 for actuating the second pilot valve **42**. By switching the second pilot valve **42** into an open position **42A**, the control line **36** can be pneumatically connected to the air release line **44** for releasing air from the control port **18.3**.

(16) The control line **36** can additionally be pneumatically connected to an emergency release port **84** and/or to an anti-compound port **82**. Via the emergency release port **84**, an emergency release pressure pSN can be provided at the control line **36** and in particular at the control port **18.3**, in order to output a parking brake pressure pBP at the spring brake port **21** independently of the pilot valve unit **32**. In particular, via the emergency release port **84**, movability of the commercial vehicle **201** can be established in a controlled manner in the event of a failure of the brake system **202** or a similar fault which leads to engagement of the parking brake function FFS. There is advantageously arranged at the emergency release port **84** a check valve **85** which opens in the direction of the control port **18.3** and closes in the opposite direction. Via the check valve **85**, an uncontrolled escape of the parking brake control pressure pPS via the emergency release port **84** can advantageously be avoided. Via the check valve **85**, a one-time inputted emergency release pressure pSN can also permanently be trapped or held in the control line **36** for permanently admitting air to the spring brake port **21** and thus for releasing at least one parking brake cylinder.

(17) Via the anti-compound port **82**, an additional parking brake pressure pSZ can be provided at the control line **36** and in particular at the control port **18.3** in order to output a parking brake pressure pBP at the spring brake port **21** independently of the pilot valve unit **32**. In particular, the additional parking brake pressure pSZ can be provided by a service brake function (not shown here), in order to implement an anti-compound function.

(18) The electropneumatic valve assembly **1** further advantageously has a selection valve **88** which is arranged in the control line **36** and is in the form of a select-high valve. The selection valve **88** has a first selection valve port **88.1** which is connected to the anti-compound port **82**, a second selection valve port **88.2** which is connected to the emergency release port **84**, and a third selection valve port **88.3** which is connected to the control port **18.3**. The selection valve **88** is configured to connect to the third selection valve port **88.3** the selection valve port, from the first selection valve port **88.1** and the second selection valve port **88.2**, at which the higher pressure is present.

(19) The electropneumatic valve assembly **1** has a first pressure sensor **64**, which is pneumatically connected to the spring brake port **21**, for measuring the parking brake pressure pBP.

(20) The electropneumatic valve assembly **1** has a trailer valve unit **90** for supplying a trailer **207** of the commercial vehicle **201** with a trailer supply pressure pVA. The trailer valve unit **90** has a trailer supply valve **92**, which in the present case is in the form of a pneumatic 3/2-way valve and to which a trailer control pressure pSA can be applied via a trailer control port **92.4** for the purpose of switching.

(21) The trailer supply valve **92** has a first trailer supply valve port **92.1**, which is pneumatically connected to the supply pressure path **70**. The trailer supply valve **92** has a second trailer supply valve port **92.2**, which is pneumatically connected to a trailer supply pressure port **96**. The trailer supply valve **92** has a third trailer supply valve port **92.3**, which is pneumatically connected to the air release line **44**.

(22) In a trailer supply position **92A**, the first trailer supply valve port **92.1** is pneumatically connected to the second trailer supply valve port **92.2** so as to pneumatically connect the supply pressure path **70** to the trailer supply pressure port **96** in order to supply the trailer supply pressure pVA.

(23) In a trailer air release position **92B**, the second trailer supply valve port **92.2** is pneumatically connected to the third trailer supply valve port **92.3** so as to connect the trailer supply pressure port **96** to the air release line **44** for the purpose of releasing air.

(24) The electropneumatic valve assembly **1** has a second pressure sensor **65**, which is

pneumatically connected to the trailer supply pressure port **96**, for measuring the trailer supply pressure pVA.

(25) The trailer pilot unit **94** has in the present case a first trailer pilot valve **97** and a second trailer pilot valve **98**. The first trailer pilot valve **97** is in the form of a 2/2-way solenoid valve and is controllable by a first electronic trailer brake signal SBA1. In an open position **97A**, the first trailer pilot valve **97** connects the supply pressure path **70** to a trailer control branch **95**, which is in turn pneumatically connected to the trailer control port **92.4**. By switching the first trailer pilot valve **97** into the open position **97A**, the trailer supply valve **92** can thus be switched into the trailer supply position **92A**. In a closed position **97B**, the first trailer pilot valve **97** is correspondingly closed and the supply pressure path **70** is separated from the trailer control branch **95**.

(26) The second trailer pilot valve **98** is controllable by a second electronic trailer brake signal SBA2. The second trailer pilot valve **98** is in the form of a 3/2-way solenoid valve. In a first position **98A**, the trailer control branch **95** is pneumatically connected to the air release port **3**. In a second position **98B**, the trailer control branch **95** is pneumatically connected to the trailer supply pressure port **96** via a throttled connection. In this manner, a self-holding function of the trailer supply valve **92** is advantageously implemented, since the outputted trailer supply pressure pVA is fed back via the second trailer pilot valve **98** in the closed position **98B** in a throttled manner to the trailer control port **92.4**. In this manner a so-called “tractor protection” functionality can be implemented, in which the supply pressure path **70** at the trailer supply valve **92** is closed by automatic switching into the trailer air release position **92B** if the pressure at the trailer supply pressure port **96** falls. An uncontrolled pressure loss via the trailer supply pressure port **96** is thereby advantageously prevented if, for example, the trailer **207** becomes detached from the commercial vehicle **201** in an uncontrolled manner and a pneumatic line connected to the trailer supply pressure port **96** breaks away.

(27) The electropneumatic valve assembly **1** has an electronic control unit **99**, which is connected in a signal- and/or energy-carrying manner to the electronic components of the electropneumatic valve assembly **1**. In particular, the first pilot valve **41**, the second pilot valve **42**, the first trailer pilot valve **97** and the second trailer pilot valve **98** are in the present case electrically connected to the electronic control unit **99** for actuation. The first pressure sensor **64** and the second pressure sensor **65** are also electrically connected to the electronic control unit **99**.

(28) FIG. 2 shows a second embodiment of an electropneumatic valve assembly **1'**. This second embodiment differs from the first embodiment shown in FIG. 1 in that a deactivation valve **60** is additionally arranged in the compensation path **86**. The deactivation valve **60** is in the present case in the form of a 2/2-way valve **61** in the form of a solenoid valve. The deactivation valve **60** is arranged between the compensation valve **80** and the control port **18.3**. The deactivation valve **60** has a first deactivation valve port **60.1**, which is pneumatically connected to the second compensation valve port **80.2**. The deactivation valve **60** has a second deactivation valve port **60.2**, which is pneumatically connected to the control port **18.3** of the main valve unit **34**, in the present case via the control line **36** and the shuttle selection valve **88**. In an activation position **60A**, the first deactivation valve port **60.1** and the second deactivation valve port **60.2** are pneumatically connected. In a deactivation position **60B**, the first deactivation valve port **60.1** and the second deactivation valve port **60.2** are separated. The deactivation valve **60** can be switched via a deactivation valve signal SD, advantageously via the electronic control unit **99**.

(29) Via the deactivation valve **60**, the functioning of the compensation valve **80** can advantageously be suppressed in a controllable manner. By switching the deactivation valve **60** into the deactivation position **60B**, the compensation path **86** is blocked, whereby in particular the control behavior of the pneumatically actuatable main valve **18** is improved because no air can escape at the control port **18.3** via the compensation valve **80**. This is advantageous in particular where time-critical control behavior is required, for example in the case of stepped actuation.

(30) FIG. 3 shows a third embodiment of an electropneumatic valve assembly **1''**. This third

embodiment differs from the first embodiment shown in FIG. 1 in that the compensation control path **83** is in the form of a feedback path, which pneumatically connects the compensation valve control port **80.4** to the spring brake port **21**. Unlike the first and second embodiments shown in FIG. 1 and FIG. 2, the parking brake pressure pBS outputted by the main valve unit **34** and not the parking brake control pressure pPS is in the present case thus provided as the compensation valve control pressure pKS at the compensation valve control port **80.4**.

(31) FIG. 4 shows an electropneumatic brake system **202**. The electropneumatic brake system **202** is in the present case used in a commercial vehicle **201**, which is here shown diagrammatically, in particular with two front wheels **212** of a front axle **210** and four rear wheels **222** of a rear axle **220**.

(32) A central control unit **700** is connected in a signal-carrying manner to an electronic control unit **99** of the electropneumatic valve assembly **1** via a central control line **310**. The electronic control unit **99** is further supplied with electric power from a power supply **710** via a supply line **320**.

(33) Two service brake cylinders **56** of a service brake function FBS, which are each associated with a front wheel **212** of the front axle **210**, can be actuated via the front axle modulator **720** for braking actuation. The front axle modulator **720** is in turn connected in a signal-carrying manner to the central control unit **700** in order to receive brake signals, wherein the central control unit **700** is able to receive brake request signals via a vehicle bus **714**. Alternatively or additionally, preferably within the context of a fallback level, the service brake function FBS can be actuated via a brake signal transmitter **718**. Compressed air from a second compressed air supply **7** is admitted to the service brake cylinders **56** via a front axle modulator **720**, in that the front axle modulator **720** provides a front axle brake pressure pBV.

(34) In an analogous manner, two parking brake cylinders **54**, which are each associated with the rear wheels **222** of the rear axle **220**, can have a service brake chamber **58** which, under the control of the brake signal transmitter **718**, can be supplied with compressed air from a first compressed air supply **6** via a rear axle modulator **722** for the purpose of braking. For this purpose, the rear axle modulator **722** provides a rear axle brake pressure pBH.

(35) The brake system **202** further has a trailer control valve **724** for providing a trailer brake pressure pBA, which can be pneumatically connected to a trailer **207** (not shown here) of the commercial vehicle **201**—for the purpose of supplying a brake system of the trailer. The trailer control valve **724** advantageously pneumatically connects the trailer supply pressure port **96** to the trailer **207** via a coupling head, of which two are shown diagrammatically as arrows on the trailer control valve **724**.

(36) The two parking brake cylinders **54** each have a spring brake and are pneumatically connected via a parking brake line **760** to a parking brake port **21** of the electropneumatic valve assembly **1**.

(37) Via the electropneumatic valve assembly **1**, a parking brake pressure pBP can be provided to the parking brake cylinders **54** in order in each case to admit air to the parking brake cylinders **54** and thus move them from a braking position **54B** into a release position **54A**.

(38) The electropneumatic valve assembly **1** is provided with a supply pressure pV, here from a third compressed air supply **8**, via a supply pressure path **70**. Equally, in other embodiments, the supply pressure path **70** can easily be connected to a different compressed air supply, in particular to the first compressed air supply **6** or to a second compressed air supply **7**. Particularly preferably, in some embodiments, two compressed air supplies can be connected to the supply pressure path **70** via a supply shuttle valve **49** as shown in FIG. 1. The electropneumatic valve assembly **1** is advantageously supplied via at least one compressed air supply, the supply pressure of which can purposively be lowered, in particular via a valve of the service brake function FBS. Such a valve can be an ABS valve **744**, for example.

(39) Via the rear axle modulator **722**, the rear axle brake pressure pBH can be provided as additional control pressure pZ at the anti-compound port **82** via the additional actuation line **764**, in particular within the context of an anti-compound function.

(40) It is understood that the foregoing description is that of the preferred embodiments of the

invention and that various changes and modifications may be made thereto without departing from the spirit and scope of the invention as defined in the appended claims.

LIST OF REFERENCE SIGNS (PART OF THE DESCRIPTION)

(41) **1, 1', 1"** electropneumatic valve assembly **3** air release port **6** first compressed air supply **7** second compressed air supply **8** third compressed air supply **18** main valve **18.1** supply port of the main valve **18.2** main port of the main valve **18.3** control port of the main valve **20** relay valve **21** spring brake port **24** parking brake valve unit **27** first throttle **28** second throttle **32** pilot valve unit **34** main valve unit **36** control line **41** first pilot valve **41A** open position of the first pilot valve **41B** closed position of the first pilot valve **42** second pilot valve **42A** open position of the second pilot valve **42B** closed position of the second pilot valve **44** air release line **49** supply shuttle valve **54** parking brake cylinder **54A** release position of the parking brake cylinder **54B** braking position of the parking brake cylinder **56** service brake cylinder **58** service brake chamber **60** deactivation valve **60A** activation position of the deactivation valve **60B** deactivation position of the deactivation valve **60.1** first deactivation valve port **60.2** second deactivation valve port **61** 2/2-way valve **64** first pressure sensor **65** second pressure sensor **70** supply pressure path **74** supply branch of the supply pressure path **76** control branch of the supply pressure path **80** compensation valve **80A** first switch position of the compensation valve **80B** second switch position of the compensation valve **80.1** first compensation valve port **80.2** second compensation valve port **80.3** third compensation valve port **80.4** compensation valve control port **81** 3/2-way valve **82** anti-compound port **83** compensation control path **84** emergency release port **85** check valve **86** compensation path **88** selection valve **88.1** first selection valve port **88.2** second selection valve port **88.3** third selection valve port **90** trailer valve unit **92** trailer supply valve **92A** trailer supply position **92B** trailer air release position **92.1** first trailer supply valve port **92.2** second trailer supply valve port **92.3** third trailer supply valve port **92.4** trailer control port **94** trailer pilot unit **95** trailer control branch **96** trailer supply pressure port **97** first trailer pilot valve **97A** open position of the first trailer pilot valve **97B** closed position of the first trailer pilot valve **98** second trailer pilot valve **98A** first position of the second trailer pilot valve **98B** second position of the second trailer pilot valve **99** electronic control unit **201** commercial vehicle **202** electropneumatic brake system **207** trailer **210** front axle **212** front wheel **220** rear axle **222** rear wheel **310** central control line **320** supply line **700** central control unit **710** power supply **714** vehicle bus **718** brake signal transmitter **720** front axle modulator **722** rear axle modulator **724** trailer control valve **744** ABS valve **760** parking brake line **764** additional actuation line FBS service brake function FFS parking brake function pBP parking brake pressure pKS compensation valve control pressure pKSS compensation valve threshold value pPS parking brake control pressure pSA trailer control pressure pSN emergency release pressure pSZ additional parking brake pressure pV supply pressure pVA trailer supply pressure SD deactivation valve signal SFB parking brake signal SBA electronic trailer brake signal SBA1 first electronic trailer brake signal SBA2 second electronic trailer brake signal SFB electronic parking brake signal SFB1 first electronic parking brake signal SFB2 second electronic parking brake signal

Claims

1. An electropneumatic valve assembly for actuating a parking brake function of an electropneumatic brake system of a commercial vehicle, the electropneumatic valve assembly comprising: at least one pressure supply; a supply pressure path; a parking brake valve unit receiving supply pressure (pV) from said at least one pressure supply via said supply pressure path; a pneumatically self-holding trailer valve unit being connected to said supply pressure path for receiving said supply pressure (pV) and having a pneumatically switchable trailer supply valve providing a trailer supply pressure (pVA) to a trailer supply port in dependence upon an electronic trailer braking signal (SBA); said parking brake valve unit including a pilot valve unit and a main

valve unit; said pilot valve unit being provided to output a parking brake control pressure (pPS) at said main valve unit; said main valve unit being configured to output a parking brake pressure (pBP) at least at one spring brake port in dependence on said parking brake control pressure (pPS) received from said pilot valve unit; and, a compensation valve being provided to maintain said parking brake control pressure (pPS) outputted at said main valve unit by said pilot valve unit and so compensate at least partially for a leakage of at least one of the following: said pilot valve unit and said main valve unit.

2. The electropneumatic valve assembly of claim 1, wherein: said compensation valve is configured as a 3/2-way valve having: a first compensation valve port for receiving the supply pressure (pV); a second compensation valve port whereat said parking brake control pressure (pPS) or a pressure derived therefrom can be present; and, a third compensation valve port connected to an air outlet; and, said compensation valve being switchable into a first switch position wherein said second compensation valve port is connected to said third compensation valve port and into a second switch position wherein said first compensation valve port is connected to said second compensation valve port.

3. The electropneumatic valve assembly of claim 2, wherein said compensation valve is biased into said first switch position by spring loading and switches into said second switch position in an actuated state.

4. The electropneumatic valve assembly of claim 2, wherein said compensation valve is in the form of a pneumatically switchable valve switchable via a compensation valve control pressure (pKS).

5. The electropneumatic valve assembly of claim 4, wherein said compensation valve is configured to switch when said compensation valve control pressure (pKS) exceeds a compensation valve threshold value (pKSS).

6. The electropneumatic valve assembly of claim 4, wherein said compensation valve control pressure (pKS) is a pressure present or outputted at said second compensation valve port so as to cause pneumatic self-holding for the compensation valve to be implemented.

7. The electropneumatic valve assembly of claim 4, wherein said compensation valve control pressure (pKS) is the parking brake pressure (pBP) present or outputted at said spring brake port.

8. The electropneumatic valve assembly of claim 1, further comprising: a control line; a compensation path pneumatically connecting said supply pressure path to said control line; said compensation valve being arranged in said compensation path; and, said control line being pneumatically connected to said main valve unit for providing said parking brake control pressure (pPS).

9. The electropneumatic valve assembly of claim 8, further comprising: a deactivation valve arranged in said compensation path and having first and second deactivation valve ports; and, said deactivation valve being switchable in dependence on a deactivation signal (SD) from an activation position pneumatically connecting said first deactivation valve port and said second deactivation valve port into a deactivation position pneumatically separating said first deactivation valve port and said second deactivation valve port.

10. The electropneumatic valve assembly of claim 1, further comprising: a trailer pilot unit for providing a trailer control pressure (pSA) in dependence on said electronic trailer brake signal (SBA); and, said trailer supply valve being pneumatically actuatable in dependence on a trailer control pressure (pSA).

11. The electropneumatic valve assembly of claim 8, further comprising: an emergency release port configured to be pneumatically connectable to said control line to provide an emergency release pressure (pSN).

12. An electropneumatic valve assembly for actuating a parking brake function of an electropneumatic brake system of a commercial vehicle, the electropneumatic valve assembly comprising: at least one pressure supply: a supply pressure path; a parking brake valve unit receiving supply pressure (pV) from said at least one pressure supply via said supply pressure path;

a pneumatically self-holding trailer valve unit being connected to said supply pressure path for receiving said supply pressure (pV) and having a pneumatically switchable trailer supply valve providing a trailer supply pressure (pVA) to a trailer supply port in dependence upon an electronic trailer braking signal (SBA); said parking brake valve unit including a pilot valve unit and a main valve unit; said pilot valve unit being provided to output a parking brake control pressure (pPS) at said main valve unit; said main valve unit being configured to output a parking brake pressure (pBP) at least at one spring brake port in dependence on said parking brake control pressure (pPS) received from said pilot valve unit; a compensation valve being provided to maintain said parking brake control pressure (pPS) outputted at said main valve unit by said pilot valve unit and so compensate at least partially for a leakage of at least one of the following: said pilot valve unit and said main valve unit; said compensation valve being configured as a 3/2-way valve having: a first compensation valve port for receiving the supply pressure (pV); a second compensation valve port whereat said parking brake control pressure (pPS) or a pressure derived therefrom can be present; and, a third compensation valve port connected to an air outlet; said compensation valve being switchable into a first switch position wherein said second compensation valve port is connected to said third compensation valve port and into a second switch position wherein said first compensation valve port is connected to said second compensation valve port; wherein said compensation valve is throttled so as to cause at least one of the following to apply: i) the connection of said first compensation valve port to said second compensation valve port is throttled; and, ii) the connection to said second compensation valve port to said third compensation valve port is throttled.

13. An electropneumatic brake system for a commercial vehicle, the electropneumatic brake system comprising: an electropneumatic valve assembly including: at least one pressure supply; a supply pressure path; a parking brake valve unit receiving supply pressure (pV) from said at least one pressure supply via said supply pressure path; a pneumatically self-holding trailer valve unit being connected to said supply pressure path for receiving said supply pressure (pV) and having a pneumatically switchable trailer supply valve providing a trailer supply pressure (pVA) to a trailer supply port in dependence upon an electronic trailer braking signal (SBA); said parking brake valve unit including a pilot valve unit and a main valve unit; said pilot valve unit being provided to output a parking brake control pressure (pPS) at said main valve unit; said main valve unit being configured to output a parking brake pressure (pBP) at least at one spring brake port in dependence on said parking brake control pressure (pPS) received from said pilot valve unit; and, a compensation valve being provided to maintain said parking brake control pressure (pPS) outputted at said main valve unit by said pilot valve unit and so compensate at least partially for a leakage of at least one of the following: said pilot valve unit and said main valve unit; at least a first compressed air supply; at least one spring brake cylinder pneumatically connected to said at least one spring brake port for receiving the parking brake pressure (pBP); and, an electronic control unit configured to provide at least one electronic parking brake signal (SFB).

14. A commercial vehicle having an electropneumatic valve assembly as claimed in claim 1.

15. A commercial vehicle having an electropneumatic brake system as claimed in claim 13.
